L3. Observation and Measurement

Observation and Measurement

Empiricism and Types of Observation

Empiricism

- **Empiricism**: the view that **sensory experience** is the ultimate source of knowledge.
- In the end, scientific claims must connect back to **what we can observe** with our senses, sometimes helped by instruments.

Direct Observation

- You use **only your own senses** to experience the event or object.
- **Examples:**
  - Seeing that a car is moving and not standing still.
  - Feeling that one bag is heavier than another.
  - Hearing that one sound is louder than another.
- **Limitations of direct observation**:
  - It can be **mistaken**:
    - A color blind person can misjudge colors.
    - Strange lighting can change how colors or shapes appear.
  - It is usually **qualitative**, not numerical:
    - You can tell "this is warmer than that", but not exactly *how much* warmer.
    - You can tell "this bag feels heavier", but not its exact weight in kilograms.

Aided Direct Observation

- You still perceive the **thing itself**, but with help from an instrument.
- **Examples**:
  - Glasses or contact lenses to see more sharply.
  - A microscope to see cells or bacteria.
  - A telescope to see distant planets or moons.
- These observations often depend on **background theory**:
  - To trust a telescope, we rely on a theory of **optics** that tells us how lenses bend light.
  - In Galileo's time, some scholars thought telescopes produced illusions, because they did not yet trust the theory.
- **Key idea:**
  - Aided direct observation is still "looking at the object", but the reliability of what you see depends on a theory of how the instrument works.

Indirect Observation

- You do **not** observe the phenomenon itself.

  Instead, you observe an **effect** that the phenomenon causes.
- General structure:

> Phenomenon (hidden or hard to observe)
>
> → causes some effect
>
> → we observe the effect and infer the phenomenon

- Examples:
  - We do not see **gravity**, but we see planets move in orbits and objects fall.
  - We do not see **elementary particles**, but we see **tracks** in a bubble chamber.
  - We do not see **temperature itself**, but we see mercury rise in a thermometer.

- **Indirect observations are always theory dependent**:
  - We need a theory that explains how the phenomenon produces the observable effect.
  - Without that theory, the effect would just be "bubbles in liquid" or "lines on a screen" and would not count as evidence for particles or fields.

Theory Dependence of Observation

- An observation is **theory dependent** if its **interpretation** depends on some accepted theory.
- Example with a microscope:
  - A scientist says "I see bacteria".
  - Strictly speaking, they see tiny shapes or dots in a microscope image.
  - It is the **theory of optics** plus **cell theory** and **staining techniques** that justifies calling those dots "bacteria".
- This shows:
  - Many observations, especially aided and indirect ones, are **not neutral**.
  - They already presuppose some theoretical assumptions in order to count as observations *of* something specific.

Logical Empiricism and Circularity

Logical Empiricism

- **Logical empiricism** is a philosophical movement with roughly these ideas:
  - Meaningful statements are either:
    - **empirically testable**, or
    - **logically or mathematically true**.
  - Metaphysical statements that cannot be tested are seen as **meaningless** for science.
  - Philosophy should clarify scientific language and logic, not introduce new untestable claims.
- Examples of how a logical empiricist classify statements:

- "Electrons have negative charge" → meaningful, because it is testable in principle.
- "God exists beyond space and time" → not meaningful for science, because it is not testable.
- "2 + 2 = 4" → meaningful, because it is analytically or logically true.

- Problem highlighted in the chapter:
  - Logical empiricism needs a **clear distinction** between:
    - observational sentences, and
    - theoretical sentences.
  - If many observations are **theory dependent**, this clear separation becomes difficult or impossible.

Threat of Circularity in Indirect Observation

- In indirect observation, we:
  1. Use a **theory** to interpret what we see.
  2. Then use that interpreted observation as **support** for the same theory.
- This risks a kind of **circularity**.

Example from particle physics:

- We see **bubble tracks** in a chamber.
- Theory says:
  - particles cause ionization in the liquid,
  - ionization causes bubbles,
  - the shape of the track is related to charge, mass, and momentum of the particle.
- So we say "We observed a particle with these properties".
- But how do we know the theory about ionization and track shapes is correct?
  - Because it fits the bubble observations.

So:

- The **theory** is used to interpret the observation.
- Then the **interpreted observation** is used to support the theory.
- This is the **threat of circularity** in indirect observation.

The main lesson:

Observation and theory support each other, rather than having a simple one way order where observation comes first and theory comes later.

Operationalization and Operationalism

Operationalization

- **Operationalization** is the process of turning an **abstract or unobservable property** into something we can **measure**.
- Examples of properties of interest:
  - Temperature
  - Stress
  - Intelligence
  - Political attitude
  - Acidity
- Examples of operationalizations:
  - Temperature → height of mercury in a thermometer.
  - Stress → self report questionnaire and cortisol level in saliva.
  - Intelligence → score on an IQ test.
  - Acidity → color of pH paper.
- Basic structure of an operationalization:
  1. **Property of interest**
     - What we want to measure, for example temperature or happiness.
  2. **Hypothesized causal or correlational link**
     - A theory that says changes in the property lead to changes in some observable effect.

Example: higher temperature leads to more expansion of a liquid in a thermometer.

3. **Directly observable effect**
    - What we actually look at, for example the height of a liquid column, the color of a strip, or answers on a survey.

A **good** operationalization has:

- A **clear property of interest**.
- A **well justified and sufficiently stable relationship** between the property and the effect.
- An **effect that is easy enough to observe with sufficient precision**.

Operationalism (too strong)

- **Operationalism** is a stronger and more problematic claim:
    - It says that a scientific concept **is nothing more than** the operation or measurement used to detect it.
- Examples of operationalism:
    - "Temperature is what the thermometer reads."
    - "Intelligence is the IQ test score."
    - "Happiness is the number you give on a 0 to 10 scale."
- Problems with operationalism:
    1. **Concept understanding without simple measurement**
        - We can understand concepts even when measurement is very complex or not yet possible, for example extremely small lengths or distances of light years.
    2. **Multiple measurements for the same concept**
        - Different thermometers or sensors can all measure temperature. If operationalism is correct, each method would define a different "temperature", which is not how we use the concept in practice.
    3. **We want to criticize measurements**

    - In real science, we often ask "Is this test a good measure of intelligence?"
      If "intelligence = IQ score", such criticism makes no sense.

- Conclusion:
    - **Operationalization** is necessary and good.
      It is about making concepts measurable.
    - **Operationalism** is too extreme and is rejected.
      It wrongly identifies the concept with a single measurement method.

Measurement: Steps, Units, and Scales

Steps of Measurement

To measure a property, we typically go through four steps:

1. **Define the concept to be measured**
    - Be clear about what exactly you want to measure.
    - Examples:
        - Temperature = degree of thermal energy of a system.
        - Stress = level of physiological and psychological strain.
        - Intelligence = ability to reason, solve problems, and learn.
2. **Specify the measurement operation**
    - Decide how the property will be detected in practice.
    - Examples:
        - Acidity → use pH paper and see the color change.
        - Stress → use a questionnaire plus cortisol analysis.
        - Length → place a ruler next to the object and see where it ends.
3. **Specify a measure and a unit**
    - Choose the instrument and the unit or scale.
    - Examples:
        - Temperature → thermometer, unit: Celsius, Fahrenheit, or Kelvin.
        - Weight → scale, unit: kilogram or pound.

    - Time → clock, unit: seconds, minutes.

4. **Represent results with numbers**
    - Record the outcome as a number, for example:
        - 37.2 °C
        - 1.5 kg
        - pH 4
        - stress score 32 out of 40

Requirements on a Good Measure

Two important criteria:

1. **Comparability**
    - The measure must have the **same type of property** as the thing being measured.
    - Examples:
        - A ruler has length. It is comparable to the length of a table.
        - A volume measure (like a measuring cup) has volume. It is comparable to the volume of a liquid.
    - A thermometer cannot directly measure length, because temperature and length are different types of properties.
2. **Stability**
    - The unit must be **sufficiently stable** over time and under normal conditions.
    - Examples:
        - A block of ice is not a stable unit of volume, because it melts and changes shape its not also satisfy comparability as ice and liquid water are not the same.
        - The oscillation of a cesium atom is extremely stable, which is why it is used to define the second.

It is also important that the measurement procedure is **public and repeatable**, so that other people can reproduce the result.

Scale Types

A **scale type** describes what kind of information the numbers carry and which mathematical operations make sense for those numbers.

Another way to define a scale type is by the **transformations of the numbers that leave the meaning of the measurement unchanged**.

Nominal Scale

- **What it is**:
  - Pure **categorization**. The numbers or labels just name different groups.
  - There is no natural order.
- **Examples**:
  - Student ID numbers.
  - Zip codes.
  - Shirt colors (red, blue, green).
  - Blood types, ethnic groups.
- **Allowed transformations:**
  - Any relabeling that preserves **non identity**.
  - For example, if dog = 1 and cat = 2, you can switch to dog = A and cat = B.
    The structure is unchanged as long as different categories stay different.

Ordinal Scale

- **What it is:**
  - A scale that represents **order** or **ranking**.
  - We know which is more or less, but not by how much.
- **Examples**:
  - Race positions: 1st, 2nd, 3rd.
  - TERRIBLE, BAD, NEUTRAL, GOOD, GREAT.
  - Movie ratings with stars.

    - Mohs hardness scale.
- We cannot say that the distance between "BAD" and "NEUTRAL" is equal to the distance between "NEUTRAL" and "GOOD".
- **Allowed transformations:**
    - Any **positive monotone** transformation, which keeps the order.
    - **Example**:

      if ranks are 1, 2, 3, we can transform them to 10, 20, 30 as long as 10 < 20 < 30.

Interval Scale

- **What it is:**
    - A scale where **equal numerical differences** represent equal differences in the property.
    - Zero is **not a true absence** of the property. It is chosen by convention.
- **Examples:**
    - Temperature in Celsius or Fahrenheit.
    - Calendar years (1000, 1100, 1200).
- We can meaningfully talk about **differences**:
    - 20 °C is 10 degrees warmer than 10 °C.
- We cannot say:
    - 20 °C is twice as hot as 10 °C, because 0 °C is not "no temperature".
- **Allowed transformations:**
    - **Positive linear** transformations:
      $x' = a\,x + b$, with $a > 0$.
    - Example: Fahrenheit and Celsius:
      $F = 1.8\,C + 32$.

Ratio Scale

- **What it is:**
    - A scale with equal intervals and a **true zero**.

- Zero corresponds to a complete absence of the property.
- Ratios are meaningful.

- **Examples**:
  - Length in meters.
  - Weight in kilograms.
  - Age in years.
  - Temperature in Kelvin.
- **We can say:**
  - 2 meters is twice as long as 1 meter.
  - 80 K is twice the absolute temperature of 40 K.
- **Allowed transformations:**
  - **Positive scalar** transformations:
    $x' = a\,x$, with $a > 0$.
  - Example: meters to inches:
    inches = 39.37 × meters.

Absolute Scale

- **What it is:**
  - A scale with a **natural zero** and a **natural unit** that is fixed by the world, not defined by convention.
- **Examples**:
  - Counting objects: 0, 1, 2, 3 apples.
  - Probability: from 0 to 1.
- We cannot rescale these values meaningfully.
  If you have 3 apples, you simply have 3 apples.
- **Allowed transformations:**
  - No non trivial transformations are allowed.
    Any change of the numbers changes the meaning.

Scale Type and Transformations: Summary

- **Nominal**: any transformation that preserves which items are equal or different.
- **Ordinal**: any transformation that preserves the order.
- **Interval**: positive linear transformations (x' = a x + b).
- **Ratio**: positive scalar transformations (x' = a x).
- **Absolute**: no non trivial transformations allowed.

Error, Accuracy, Precision, and Validity

Systematic Error (Bias)

- A **systematic error** is a **consistent, repeatable** error that shifts all measurements in the **same direction**.
- It affects **accuracy**, that is how close the average measurement is to the true value.
- **Examples**:
  - A scale that is miscalibrated and always reads +2 kg too high.
  - A ruler that is slightly too short, so all length measurements are too small.
- A series of measurements can be:
  - **Precise but biased**.
    The results are close to each other but far from the true value.

Random Error (Noise)

- A **random error** causes measurements to **fluctuate unpredictably** around the true value.
- It affects **precision**, that is how tightly repeated measurements cluster together.
- **Example**:
  - You weigh yourself several times and get slightly different numbers each time, even though your real weight is not changing.
- **Important property:**

- If there is **no systematic error**, the average of many repeated measurements can reduce the impact of random error and get closer to the true value.

Precision vs Accuracy

- **Precision**:
  - Low random error.
  - Repeated measurements cluster tightly.
- **Accuracy**:
  - Low systematic error.
  - Measurements are close to the true value.
- You can have:
  - High precision and low accuracy (clustered but biased).
  - Low precision and reasonable accuracy on average (values scattered around the true value).

Convergent and Divergent Validity

These concepts help us judge whether a measurement method is working properly, and they can also hint at **systematic errors**.

Convergent Validity

- **Convergent validity** is achieved when **different, independent methods** for measuring the **same property** give **similar results**.
- **Example**:
  - Three different thermometers (mercury, gas, infrared) give nearly the same temperature for the same object.
- If they do **not** converge, at least one method may contain **systematic error**.

Divergent Validity

- **Divergent validity** is defined as follows:
  A method has divergent validity if it gives **different results** when applied to **substantially different targets i**f we apply a test on A and another one to B and A and B are **different** then we should observe **different measurements**.
- **Example:**
  - A thermometer should give a low value for and ice cube and a high value for boiling water. If it doesn't, there is an issue because obviously the ice cube is colder.
  - If we measure someones heartbeat when they are running and when they are resting, we should get different results.
- If a measurement method gives almost the **same result for clearly different cases**, it may be insensitive or systematically flawed.
